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MILITARY SPECIFICATION

TELESCOPE, ELBOW: M93

1. SCOPE

1.1 This specification covers one type of elbow telescope known as "Telescope, Elbow: M93", hereinafter referred to as "telescope" (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-O-13830—Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of.

MIL-F-13926—Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL-P-14232—Parts, Equipment and Tools for Ordnance Materiel, Packaging of.

MIL-I-45208—Inspection Requirements: General Specification for.

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109—Quality Assurance Terms and Definitions.

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F7653039—Telescope, Elbow: M93.

IEL-7653039—Index of Inspection Equipment Numbers for Telescope, Elbow: M93.

LSQAP-7653039—List of Supplementary Quality Assurance Provisions for Telescope, Elbow: M93.

PACKING DATA SHEET

MIL-P-14232/P7653039—Packaging of Telescope, Elbow: M93.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

FSC 1240

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NATIONAL BUREAU OF STANDARDS
NBS Resolution Test Chart of 1952
(High Contrast) (Number 25X)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Fabrication. The telescope shall be manufactured in accordance with Drawing F7653039, "Telescope, Elbow: M93", and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure inherent quality and performance capabilities.

3.2 General specifications. The contractor shall be responsible for adherence to and compliance with the following requirements of Specifications MIL–O–13830 and MIL–F–13926.

3.2.1 *MIL–O–13830.*

(a) Cleanliness.
(b) Illuminated reticle.

3.2.2 *MIL–F–13926.*

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Electrical and electronic assemblies.
(e) Part identification and marking.
(f) Workmanship.

3.3 Environmental.

3.3.1 *Storage temperatures.* The telescope shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of 60° to 90° F. after having been exposed to and thermally stabilized at ambient temperatures of +160° and —80° F.

3.3.2 *Operating temperatures.* The telescope shall meet the applicable requirements of 3.6.12 while exposed to and thermally stabilized at ambient temperatures of +150° and —40° F., subsequent to which the telescope shall meet the requirements of 3.3.3 to 3.6.13, inclusive, when thermally stabilized at a standard ambient temperature of 60° to 90° F.

3.3.3 *Vibration.* The telescope shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of $\frac{1}{16}$ inch ($\frac{1}{8}$ inch maximum excursion) for a period of 5 minutes ±15 seconds. Subsequent to vibration, the telescope shall show no evidence of loose or damaged parts and shall be capable of meeting the requirements of 3.4 to 3.6.13 inclusive.

3.4 Cleanliness. Subsequent to vibration, the telescope shall meet the cleanliness requirements of Specification MIL–O–13830. The surfaces of the reticle shall show no more than five particles of dirt. The area of each particle shall not be in excess of the area equivalent to a size five dig. There shall be no other dirt within the telescope which shall adversely affect observation and image quality.

3.5 Sealing. The telescope shall show no evidence of leakage when subjected to an external pressure of dry nitrogen at five pounds per square inch gage pressure (psig) for a period of at least five minutes. Dew point of nitrogen shall be at least as low as —25° F.

3.6 Accuracy.

3.6.1 *Collimation.* The telescope shall be mounted on the spherical surfaces of the mounting bearings and the flat surface of the elbow with the telescope horizontal. The line of sight through the center of the boresight cross shall be 40 mils ± 1.0 mil above the geometric axis of the telescope and shall coincide within 0.1 mil of a plumb line passing through the geometric axis when the azimuth and elevation boresight knobs are each in coincidence with their zero indexes.

3.6.2 *Collimation retention.* The line of sight through the boresight cross shall not deviate from target point by more than 0.25 mil when the cant correction coupling is rotated to move the eyepiece through an excursion of 600 mils. Movement shall be 300 mils clockwise and 300 mils counter-

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clockwise from the position established in 3.6.1.

3.6.3 *Boresight travel.* Boresight knobs shall be capable of moving the image ± 10 mils in azimuth and elevation from their zero index positions. The field of view shall be circular when the boresight knobs are at zero index setting (see 6.2).

3.6.4 *Parallelism of reticle and image.* With the telescope positioned to the requirements of 3.6.1, and the eyepiece rotation scale (cant correction) set at zero index, a 60 mil horizontal reticle line shall not deviate from a horizontal target line by more than 0.35 mil (block) at one extremity of the reticle line when the other extremity of the reticle line is in coincidence with the horizontal target line.

3.6.5 *Image tilt.* The image of a plumb line shall be vertical within one degree (one degree is equivalent to 3.2 mils at the edge of the field when measured across the full field of view with the boresight knobs at zero index settings).

3.6.6 *Eyepiece focus.* The eyepiece focus shall be fixed between the limits of —0.75 and —1.0 diopter.

3.6.7 *Resolution.* The resolution on the optical axis shall be twelve seconds of arc or less.

3.6.8 *Parallax.* Parallax at the geometric axis shall not exceed 0.12 mil (reticle line width) when viewing a target at 300 yards ± 10 yards, when the boresight knobs are set at their respective zero index.

3.6.9 *Reticle accuracy.* The angular value for reticle graduations between zero and maximum range shall read as indicated on the reticle drawing within 0.25 mil.

3.6.10 *Reticle gearing.* The gearing between the cant correction coupling and the rotation of the reticle shall be accurate to ± 2.0 mils, including backlash, when the reticle is rotated 300 mils clockwise or counterclockwise of its central position (one revolution of the coupling equals 59.26 mils rotation of the reticle).

3.6.11 *Boresight locks.* Actuation of the boresight locks shall not cause a deflection of the line of sight. When the boresight locks are secured, application of a torque of 10 pound inches at standard ambient temperature of 60° to 90° F. shall not rotate the knobs.

3.6.12 *Operability.*

3.6.12.1 *Boresight adjustment.* The running torque required to rotate the boresight adjustments with the locks released shall not exceed 5 pound inches at a standard ambient temperature of 60° to 90° F. The values obtained at standard ambient temperature shall not be exceeded by more than 3 pound inches when the adjustments are checked at the extreme temperatures of 3.3.2.

3.6.12.2 *Cant correction coupling.* The running torque required to rotate the cant correction coupling shall not exceed 5 pound inches at a standard ambient temperature of 60° to 90° F. The value obtained at standard ambient temperature shall not be exceeded by more than 10 pound inches when the adjustments are checked at the extreme temperatures of 3.3.2.

3.6.13 *Reticle illumination.* With the reticle illuminated, the reticle markings shall be clearly defined when observed in a darkened area.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections

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are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL–I–45208 and the component inspection requirements of Specification MIL–F–13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL–STD–109.

4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" specified in Standard MIL–STD–105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL–F–13926 and List of Supplementary Quality Assurance Provisions, LSQAP–7653039. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL–STD–105 and the inspection level and sampling plans specified in Table I titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL–P–14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE I. *Classification of Defects*
Use Level II of Table I with Sampling Plan Table II–A of Standard MIL–STD–105

	Requirement	*Test procedure*
Critical: None.		
Major: AQL 1.5%		
101. Vibration	3.3.3	4.5.3
102. Sealing	3.5	4.5.5
103. Collimation	3.6.1	4.5.7
104. Collimation retention	3.6.2	4.5.8
105. Parallelism of reticle and image	3.6.4	4.5.12
106. Eyepiece focus	3.6.6	4.5.14
107. Resolution	3.6.7	4.5.15
108. Parallax	3.6.8	4.5.13
109. Reticle accuracy	3.6.9	4.5.11
110. Boresight locks	3.6.11	4.5.17
Minor: AQL 2.5%.		
201. Cleanliness	3.4	4.5.4
202. Boresight travel	3.6.3	4.5.9
203. Image tilt	3.6.5	4.5.10
204. Reticle gearing	3.6.10	4.5.16
205. Operability (60° to 90° F.)	3.6.12	4.5.18
206. Reticle illumination	3.6.13	4.5.19

4.2.3 *Disposition of non-conforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL–STD–105.

4.3 Control tests.

4.3.1 *General.* One telescope shall be selected at random from each 100 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

Control test	*Requirement*	*Test Procedure*
Fabrication	3.1	Drawings and Specifications
General Specifications	3.2	Visual and Drawings

4.3.2 *Storage and operating temperatures.* Three telescopes shall be selected at random from each 50 produced, or from each month's production, whichever occurs first, to determine compliance with the storage and operating temperature requirements and tests specified in 3.3.1, 3.3.2, 4.5.1, and 4.5.2 respectively.

4.3.3 *Control test failure.* Should the con-

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trol test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 **Inspection equipment.** Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Inspection Equipment" requirement of Specification MIL–I–45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection equipment.* Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Inspection Equipment", and "Government Furnished Material" requirements of Specification MIL–I–45208.

4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Government design.* All inspection equipment required by drawings forming a part of the contract and not provided by the Government shall be supplied by the contractor in accordance with the Index of Inspection Equipment Numbers, IEL–7653039. Alternate designs may be substituted only as provided by "Inspection Provisions" of Specification MIL–I–45208.

4.4.2.2 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.5 **Test methods and procedures.**

4.5.1 *Storage temperature.* The temperature cycling for the physical endurance test of the telescope at +160° and —80° F. shall be conducted in conformance with 3.3.1 and the "Test Facilities" of MIL–F–13926. Upon return to standard ambient temperature, the telescope shall be subjected to a visual and tactile examination and shall show no evidence of mechanical or optical failure. The telescope shall then be subjected to the tests specified herein, and shall conform to the requirements specified.

4.5.2 *Operating temperatures.* The temperature cycling for the operability tests of the telescope at the specified temperature extremes shall be conducted in conformance with 3.3.2 and the "Test Facilities" of MIL–F–13926. When the telescope is thermally stabilized at +150° and —40° F., respectively, the boresight adjustment and the cant correction coupling of the telescope shall be subjected to the tests of 4.5.18.

4.5.3 *Vibration.* Standard vibrating equipment, capable of providing the amplitude and frequency specified in 3.3.3, shall be utilized for this test. The telescope shall be vibrated in a vertical plane. The test equipment shall be in conformance with the "Test Facilities" requirement of Specification MIL–F–13926.

4.5.3.1 *Post vibration.* Subsequent to vibration the telescope shall be thoroughly examined for physical defects resulting from poor workmanship and material to determine compliance with 3.3.3 prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts prior to being subjected to the remaining performance tests. Minimum visual examination shall include the housing, eyeshield, nuts, bolts, etc. A tactile examination shall also be conducted to determine looseness or improper function of the boresight locks, reticle gearing, azimuth and elevation boresight knobs, etc. Optical

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characteristics shall be examined by observing through the eyepiece and objective to detect excessive dirt obstructions or interference with the optical system that would be cause for rejection. An audible examination shall be made by vigorously shaking the telescope while listening for any loose internal mechanical or optical parts (lens, reticle, prisms, cells, window, etc.). Any one of the possible defects defined above shall be cause to consider the telescope as defective. Telescopes which have passed all the above examinations shall be tested for the requirements of 3.4 to 3.6.13 inclusive.

4.5.4 *Cleanliness.* Subsequent to the vibration test, the cleanliness test shall be visual in accordance with 3.4 and Specification MIL–O–13830. There shall be no visible evidence of moisture on the internal surfaces of the telescope. Dirt particles shall not exceed the tolerance as specified in 3.4.

4.5.5 *Sealing.* The telescope shall be subjected to the external pressure as specified in 3.5 for the required time period. The pressure shall be applied and measured with a fixture and air-reservoir conforming to Drawings F7695375 and C7695383, which are listed on IEL–7653039. Any pressure drop during the five minute time period shall constitute a reject.

4.5.6 *Orientation.* The following tests will be performed with the telescope orientated as referenced on Drawing F7695296 in accordance with the "set-up" and "operating" procedures therein.

(a) Parallax.
(b) Reticle accuracy.
(c) Collimation.
(d) Boresight travel.
(e) Parallelism of reticle and image.
(f) Image tilt.
(g) Collimation retention.

4.5.7 *Collimation.* With the telescope positioned as depicted on fixture Drawing F7695296, the boresight cross shall coincide with a plumb line passing through the geometric axis and shall be located 40 mils above the geometric axis within the tolerance stipulated in 3.6.1. Proper location of the boresight cross will be determined by means of a collimator reticle.

4.5.8 *Collimation retention.* With the telescope positioned as in 4.5.7, the line of sight through the boresight cross shall not deviate from a target point by more than the tolerance specified in 3.6.2 when the cant correction coupling is actuated to rotate the eyepiece through an excursion of 300 mils in each direction from the zero setting. The collimator reticle shall provide limit lines to determine accuracy.

4.5.9 *Boresight travel.* With the telescope positioned as in 4.5.7, the boresight knobs shall be capable of moving an image the distance as specified in 3.6.3 from the position outlined in 4.5.7. This test is performed by moving the knobs while simultaneously viewing the target 10 mil limit lines of the collimator reticle through the telescope's eyepiece.

4.5.10 *Image tilt.* With the fixture set up as outlined on Drawing F7695296 and with an auxiliary telescope aligned to the collimator's reticle centerline, the vertical line of the auxiliary telescope shall be within the limits of the tolerance lines provided on the collimator's reticle in accordance with 3.6.5.

4.5.11 *Reticle accuracy.* This test may be accomplished by using the set-up described in 4.5.6. The reticle range line shall fall within the limits of the tolerance lines provided on the collimator reticle in accordance with 3.6.9.

4.5.12 *Parallelism of reticle and image.* This test is accomplished by positioning one end of the telescope's 60 mil horizontal reticle line in coincidence with target line provided and observing whether the opposite end of the reticle line falls within the limits as indicated by marks on the collimator reticle.

4.5.13 *Parallax.* This test may be performed while the telescope is positioned in the test fixture as outlined in 4.5.6. Parallax is tested by coinciding a reticle line near the geometric center to the reticle mark provided

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for this test. Parallax is observed as the displacement between the reticle lines of the telescope and collimator while the observer's head is moved from side to side or up and down. The telescope must be parallax free within the limits outlined in 3.6.8.

4.5.14 *Eyepiece focus.* This test is performed with the aid of a dioptometer of at least three-power. After the dioptometer has been focused to the individual inspector's eye and the diopter scale set to zero diopters, position the dioptometer behind the telescope's eyepiece. Place a white background before the telescope's objective. Align the dioptometer's reticle cross line close to the center line of the telescope's reticle. Focus the dioptometer's objective until its reticle cross lines are in focus with the telescope's centerlines. The diopter scale reading shall be within the limits specified in 3.6.6.

4.5.15 *Resolution.* Resolution shall be tested using the number 28 test bar pattern of the National Bureau of Standards and at least a six power dioptometer. The position of the test bars on the target shall be as specified in Specification MIL–O–13830. When using this test bar pattern the target distance shall be 35.29 feet from the objective lens of the telescope under test. The target shall be properly illuminated during this test. Prior to performing this test, adjust the dioptometer to the user's individual eye accommodation, set the scale at zero, and remove parallax.

4.5.15.1 *Performing the test.* The test is performed by sighting through the telescope to the centerpoint of the test target. Next, position the dioptometer behind the eyepiece of the telescope under test and align its optical axis to the previously aligned optical axis and target centerpoint. Sight through the dioptometer and observe the test target. Acceptable resolution is assured when the test bar pattern is resolved in each of the four meridians. Resolution of the optical axis shall be as specified in 3.6.7.

4.5.16 *Reticle gearing.* Backlash between the cant correction coupling and the reticle rotation gear mechanism shall be checked, when the telescope is positioned as specified in 4.5.7, to the tolerance of 3.6.10. When this position has been obtained, scribe an index line on the coupling and note the position of the scribe line. Rotate the coupling clockwise and note the position of the reticle drift line with respect to the tolerance line on collimator reticle representing an angle of 600 mils. Note the position of the scribe line. Continue to rotate the coupling beyond the tolerance lines and return the index to the previously established position and observe where the reticle drift falls with respect to the tolerance lines on the collimator reticle. The telescope drift line must fall within the tolerance lines of the collimator reticle. Repeat this procedure by rotating the coupling counterclockwise from the zero position.

4.5.17 *Boresight locks.* With the telescope's line of sight in coincidence with a target point, actuation of the boresight locks shall not cause the line of sight to move. With the boresight locks secured, the boresight knobs shall not rotate when a torque of 10 pound inches is applied in a clockwise and counterclockwise direction by means of an appropriate adapter and torque wrench.

4.5.18 *Operability.* Running torque at standard ambient and extreme temperatures shall be measured by means of a torque wrench and adapter. During this test, the initial reading produced to start rotation of the boresight knob shall be omitted. The running torque at standard ambient and extreme temperatures shall be as specified in 3.6.12.1 and 3.6.12.2.

4.5.19 *Reticle illumination.* The inspection procedure of Specification MIL–O–13830 shall apply.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing, and marking shall be in accordance with Specification MIL–P–14232 and Packaging Data Sheet MIL–P–14232/P7653039. The level of protection

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shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Elbow Telescope M93 is primarily intended to lay the 155 mm Howitzer: M44 in direct fire.

6.2 Field of view. An elliptical field of view results from rotation of the boresight knobs. Rotation of one boresight knob moves the image in a generally vertical direction with a small horizontal component, rotation of the other boresight knob moves the image in a generally horizontal direction with a small vertical component. A true circular field of view will be obtained only when the boresight knobs are in coincidence with their zero indexes.

6.3 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable stock number.
(c) Selection of applicable levels of preservation, packaging, packing, and marking.
(d) Applicable packaging data sheet number (see 5.1).

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